



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO SWARNAMBIGAI FUEL SERVICE

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

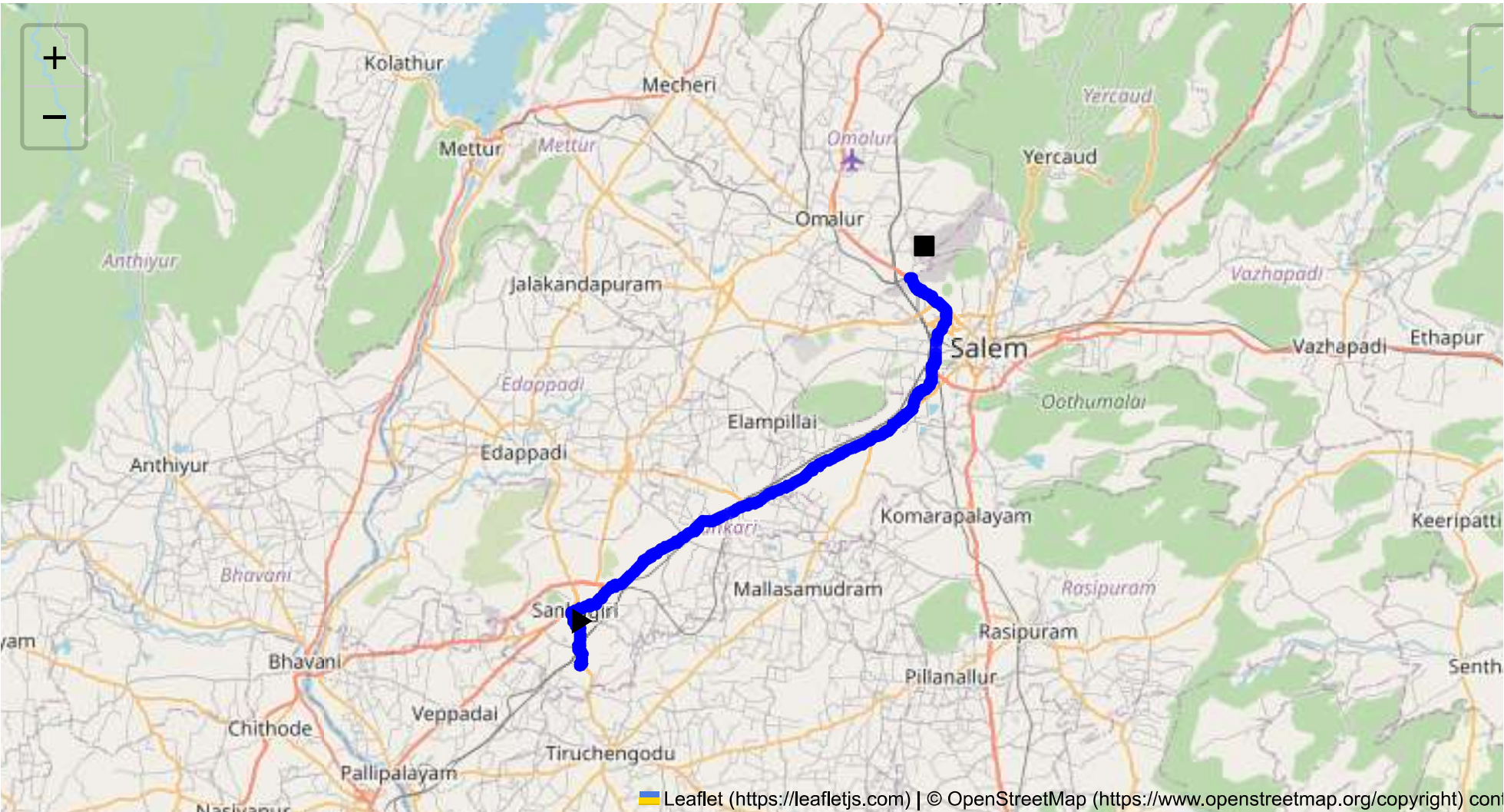
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 48.63 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.3 hours
Start: (11.4381, 77.8734)
End: (11.693353, 78.111915)



Welcome to the Journey Risk Management Study

Route Safety and Risk Assessment Report

Route Overview: The route from CVQF+23W, Sangagiri to M4V6+7X7, NH 44, Mamangam, Salem, Tamil Nadu spans approximately 48.63 kilometers. The journey primarily involves travel along national highways, with a mix of rural roads near the starting and ending points.

1. Route Map Overview:

- Starting from Sangagiri, the route likely involves NH 79 and NH 44, which are major highways facilitating efficient transportation to Salem.
- NH 44, being a part of the Golden Quadrilateral, is well-maintained and crucial for regional connectivity.

2. Weather Conditions:

- The region experiences a tropical climate, with hot summers and moderate to heavy monsoon rains.
- Summer months (April to June) can see temperatures exceeding 40°C, risking overheating of vehicles and cargo.
- Monsoon season (July to September) may cause waterlogging or reduced visibility, potentially affecting driving conditions.

3. Traffic Patterns:

- Peak hours in the morning (8:00 AM to 10:00 AM) and late afternoon (5:00 PM to 8:00 PM) see increased congestion, especially nearing Salem.
- Congestion-prone areas include junctions on NH 44 and near Salem city limits due to higher urban traffic volume.

4. Road Quality and Infrastructure:

- NH 44 is generally well-maintained with smooth road surfaces, clear signages, and good lighting, especially compared to smaller, rural roads.
- Some stretches near Sangagiri might have uneven surfaces, requiring cautious driving.

5. Alternative Routes for Emergencies:

- An alternative to NH 44 could be via SH 86, although this might increase travel time and distance.
- Local roads can provide detours around specific congestion points but may not be suitable for heavy vehicles.

6. Local Regulations on Hazardous Material Transport:

- Regulations require vehicles carrying hazardous materials to be clearly marked and equipped with requisite safety measures.
- Night-time travel for hazardous materials might be restricted or require special permits.

7. Historical Incidents:

- Historical data suggests occasional accidents involving heavy vehicles, primarily due to driver fatigue or poor weather conditions.
- Past incidents have highlighted the importance of adequate vehicle maintenance checks before travel.

8. Environmental Considerations:

- The route passes through semi-rural areas with agricultural land; care must be taken to minimize spill risks.
- Nearby water bodies and the presence of rural settlements necessitate adherence to environmental safety regulations.

9. Communication Coverage:

- Mobile network coverage is generally reliable along NH 44 and urban areas.
- Potential dead zones may exist in rural stretches, particularly at the beginnings and ends of the route.

10. Emergency Response Times:

- Emergency services in urban areas like Salem can respond within 15-30 minutes.
- In rural areas close to Sangagiri, response times may extend to 45 minutes or longer depending on the distance from emergency facilities.

11. Overall Risk Assessment:

- The route is generally safe with well-maintained infrastructure along NH 44, although minor roads require careful navigation.
- Weather conditions and traffic congestion are the primary risk factors, along with occasional communication dead zones.
- Compliance with transport regulations and regular vehicle checks can significantly mitigate risks.

Summary: Overall, the route from Sangagiri to Salem via NH 44 offers a reliable transit way for heavy vehicles carrying hazardous materials, with proper planning around peak traffic hours and weather conditions. Adhering to safety regulations and ensuring preparedness can significantly minimize potential hazards along this corridor.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43956, 77.87340	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Turn	High	11.45004, 77.87206	15 KM/Hr
5	Turn	Medium	11.45028, 77.87202	30 KM/Hr
6	Turn	High	11.47403, 77.86841	15 KM/Hr
7	Turn	Medium	11.47410, 77.86845	30 KM/Hr
8	Turn	High	11.49149, 77.89601	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
9	Turn	High	11.49224, 77.89601	15 KM/Hr
10	Turn	Medium	11.61736, 78.10588	30 KM/Hr
11	Turn	Medium	11.61753, 78.10608	30 KM/Hr
12	Turn	Medium	11.64007, 78.11794	30 KM/Hr
13	Turn	Medium	11.64020, 78.11786	30 KM/Hr
14	Turn	Medium	11.68921, 78.11798	30 KM/Hr
15	Turn	Medium	11.68926, 78.11777	30 KM/Hr
16	Turn	High	11.69390, 78.10964	15 KM/Hr
17	Turn	High	11.69394, 78.10964	15 KM/Hr
18	Blind Spot	Blind Spot	11.70185, 78.10306	10 KM/Hr
19	Turn	High	11.70200, 78.10318	15 KM/Hr
20	Blind Spot	Blind Spot	11.69310, 78.11182	10 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	police	Police Quarters	11.4741243, 77.8673495	30 km/h	Medium
1	hospital	Kathir Eye Hospital	11.4740976, 77.8681591	30 km/h	Medium
2	hospital	Thilagam Hospital	11.4753573, 77.8685829	30 km/h	Medium
3	clinic	Meyyazhagan Clinic	11.4752254, 77.8685671	30 km/h	Medium
4	hospital	Dr. Jaganathan Hospital	11.4752717, 77.8709312	30 km/h	Medium
6	hospital	Sankari Hospital	11.4777496, 77.873892	30 km/h	Medium
8	clinic	Dr. Dhanasekar Clinic	11.4793113, 77.8829247	30 km/h	Medium
9	hospital	Sri Vellaiyan Hospital	11.4798876, 77.8831394	30 km/h	Medium
11	hospital	C. N. Hospital	11.55724, 78.0096	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
12	hospital	Annapoorna Medical College Hospital	11.57693, 78.037627	30 km/h	Medium
13	hospital	Vinayaka Mission's Homeopathic Medical College & Hospital	11.583617, 78.059145	30 km/h	Medium
14	hospital	Vinayaka mission Kirubananadah variyar medical college hospital	11.584984, 78.0618417	30 km/h	Medium
15	hospital	vims	11.5854239, 78.0639188	30 km/h	Medium
18	hospital	Vinayaka Mission’s Kirupananda Variyar Medical College	11.5977669, 78.0850992	30 km/h	Medium
22	hospital	Bharathi Hospital, Salem	11.6492647, 78.1204131	30 km/h	Medium
23	hospital	Sisu Hospital Inida Ltd	11.6505276, 78.1205003	30 km/h	Medium
24	hospital	Aishwarya Hospital	11.6716291, 78.1267807	30 km/h	Medium
25	hospital	C.M. Hospital	11.6747345, 78.1255034	30 km/h	Medium
26	pharmacy	Healthy Pharmacy	11.6762714, 78.127744	30 km/h	Medium
27	pharmacy	Thulasi Pharmacy	11.6761991, 78.1277377	30 km/h	Medium
29	hospital	Sri Gokulam Speciality Hospital	11.6788724, 78.1274258	30 km/h	Medium
30	hospital	Global Medical Center	11.6867049, 78.1186697	30 km/h	Medium
31	hospital	Manipal Hospital	11.703072, 78.1026612	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
5	school	Goverment Boys Higher Secondary School	11.477659, 77.8659632	30 km/h	Medium
7	marketplace	Sunday Market	11.4796502, 77.8708458	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	university	Vinayaka Missions Vinayaka Missions Research Foundation	11.5974056, 78.0843617	30 km/h	Medium
17	school	Vinayaka Missions Annapoorani Matric School	11.5971662, 78.0839876	30 km/h	Medium
19	school	Dist Education Training	11.6133799, 78.1022409	30 km/h	Medium
20	school	DIET - Salem	11.6144158, 78.1025078	30 km/h	Medium
21	school	Diamond Rays Matric HR Sec School	11.6385629, 78.1206259	30 km/h	Medium
28	college	Sona College	11.6768, 78.1257052	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.45004, 77.87206



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45028, 77.87202



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.47403, 77.86841



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.47410, 77.86845



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49149, 77.89601



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49224, 77.89601



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.61736, 78.10588



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.61753, 78.10608



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64007, 78.11794



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.64020, 78.11786



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68921, 78.11798



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.68926, 78.11777



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69390, 78.10964



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.69394, 78.10964



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.70185, 78.10306



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.70200, 78.10318



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.69310, 78.11182

